Sahaj Bhadja

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Summary

Systems software engineer with 2+ years building **C++/Python applications**, simulation and data pipelines, observability workflows, and cross-platform releases; experienced reproducing anomalies, analyzing logs and metrics, documenting root causes, testing fixes, and shipping through CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Investigated **production data-stream anomalies** across 5,000+ monthly jobs by reconstructing job states from logs, isolating malformed inputs and failed integrations, and implementing validated fixes that achieved 99%+ reliability.
- Built **Python services and diagnostic data workflows** for 500+ users, optimizing indexed retrieval and access patterns to reduce latency from 1.2 seconds to 350 milliseconds while improving failure visibility.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Debugged **unfamiliar Python processing and integration code** under production load, reproducing failures across PDF, DOCX, and ZIP inputs and adding validation, persistent results, fallback handling, and automated tests.
- Developed **concurrent data-processing services** handling 1,000+ files per batch and 10,000+ daily requests, using structured usage logs and fault isolation to reduce turnaround time 60% while maintaining stable execution.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python simulation and evaluation pipelines** processing 100,000+ autonomous-system trajectories, integrating map geometry and temporal signals while creating reproducible closed-loop scenarios that improved long-horizon planning performance 18%.
- Analyzed **metrics, constraints, and failure patterns** across three autonomy architectures, visualizing recorded trajectory behavior and documenting root causes and corrective software changes that improved simulated closed-loop stability 15%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Diagnosed and fixed **cross-platform API and data-pipeline defects** in established codebases, applying OOP, SQL, caching, and performance analysis to improve response times 30%, pipeline efficiency 68%, and maintainability.
- Automated **versioned builds, deployments, and 20+ tests** with GitHub, Jenkins, PyTest, and Mockito, generating coverage reports and failure logs while validating releases across Windows, Linux, and macOS environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **real-time data synchronization** across eight services and 2M+ records using REST APIs, WebSockets, and Azure SQL, reducing consistency conflicts 33% while supporting authentication, billing, permissions, and notifications.
- Used **Sentry, Azure Monitor, and Kibana logs** to reconstruct failures, identify service-level causes, and verify fixes through Jest and Selenium before Docker packaging and staged Jenkins deployment.

Skills

Languages: C++, Python, Java, SQL, Bash, JavaScript, TypeScript
Systems: Linux, Windows, OOP, data structures, algorithms, multithreading, concurrency, simulation, performance analysis
Debugging: Root-cause analysis, anomaly reproduction, failure isolation, log analysis, metrics, Sentry, Azure Monitor, Kibana
Data & APIs: REST APIs, WebSockets, real-time pipelines, SQL Server, PostgreSQL, MongoDB, validation, reconciliation
Delivery: Git, Jenkins, GitHub Actions, CI/CD, Docker, PyTest, Mockito, Jest, Selenium, documentation, Agile/Scrum

Projects

CPU Scheduling Simulator Live | *C++, React.js, OS Internals*
- Implemented and tested **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and live performance metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.

PiXMod Live | *Python, PyTorch, SAM, Stable Diffusion*
- Built a **computer-vision processing pipeline** using SAM, Stable Diffusion, and PyTorch, processing 1,000+ edited assets while storing images and metadata in MongoDB for scalable retrieval, diagnostics, and analysis.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a **containerized Python processing service** on AWS ECS/Fargate with CodePipeline, automated validation checks, and versioned CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.